

Derin Önder EREN

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Education

Mount Allison University

Sackville, NB, Canada

BSc in Computer Science (Major), Mathematics & Commerce (Minors)

Sept 2024 — Present

- **Leadership & Activities:** Vice President (External) of Math & CS Society, Competitive Programming Team
- **Relevant Coursework:** Data Structures, Program Development, Microprocessors, Abstract Algebra I, Linear Algebra, Discrete Mathematics, Multivariable & Single Variable Calculus

Work Experience

Software Developer

Aug 2022 — Sep 2024

AymarTech

Istanbul, Turkey

- Engineered high-throughput, scalable backend systems with FastAPI, MongoDB, and Rust, supporting 1,000+ concurrent users with low latency.
- Implemented responsive front-end interfaces and cross-platform UI components to enhance core user workflows.
- Streamlined deployment pipelines on AWS and Docker, accelerating CI/CD efficiency and significantly reducing operational overhead.

Game Developer Intern

Feb 2024 — Jun 2024

Brilliant Labs

Moncton, NB, Canada (Remote)

- Developed a complete 2D platformer in Godot, authoring custom 2D physics, level geometry, and AI-driven NPC behaviors.
- Optimized rendering pipelines and collision detection routines to maintain steady frame rates on low-spec hardware.
- Contributed to agile iterations, participating in cross-functional feature reviews, rapid prototyping, and playtesting sessions.

Technical Projects

Rust CHIP-8 Emulator (github.com/d3r1n/rust-chip)

2024 — 2024

- Built a low-level architecture emulator from scratch using Rust, implementing complete CPU instruction cycles, memory mapping, input handling, and display rendering.
- Applied systems-level programming concepts including opcode decoding, stack management, memory-mapped I/O, and register-level debugging.

dotman — Linux Configuration Manager (github.com/d3r1n/dotman)

2024 — 2024

- Programmed a CLI tool in Rust to manage, synchronize, and deploy Linux configuration files across modular environments.
- Automated bootstrap scripting, backup workflows, and profile-based setups for cross-distribution compatibility.

Extracurriculars & Leadership

VP Workshops & Lead Web Developer — MtA Hacks

Sep 2026 — Present

- Designed, built, and deployed mtahacks.ca, delivering a high-performance, accessible, and responsive user interface.
- Led technical workshop programming and coordinated mentors for Mount Allison's flagship hackathon.
- Set-up and maintained the event's infrastructure, including CI/CD pipelines, server hosting, and deployment automation.

Honors & Competition Awards

- **2nd Place** | NCC Mastercard East Regional CTF (Flames Team): Podium finish solving multi-stage challenges in Web Exploitation, Cryptography, Reverse Engineering, and Forensics.

- **3rd Place** | Teknopark Istanbul CTF: Captained a 5-member team in executing real-world exploits under strict time constraints.

Technical Skills

- **Languages:** Rust, C, C++, Python, TypeScript, JavaScript, Go, Java, C#, Zig, Assembly, Lua, Elixir, Dart, R, HTML/CSS
- **Systems & Embedded:** Bare-metal programming, Microprocessors, Raspberry Pi, ESP32, STM32, RP2040, Arduino
- **Backend & Infrastructure:** FastAPI, Node.js, MongoDB, PostgreSQL, Redis, AWS, Docker, Podman, Linux, CI/CD (GitHub Actions), Makefiles
- **Frontend & Frameworks:** React, Svelte, SvelteKit, Vue, Solid.js, Next.js, Flutter, Tailwind CSS